Gayatri Vidya Parishad College of Engineering Autonomous

Electrical and Electronics Engineering | Course Code: EE201

Electrical Circuit Analysis - Comprehensive Syllabus Lecture Notes

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Syllabus Focus: EE201 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Electrical Circuit Analysis. Includes detaile

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Network Theorems DC & AC

" Syllabus Scope: Thevenin, Norton, Superposition, Maximum Power Transfer, Reciprocity, and Millman theor

Unit 2: Transient Response of First-Order Circuits

" Syllabus Scope: Natural and forced response of RL and RC circuits, Differential equations, Time constan

EE201: Electrical Circuit Analysis - Advanced Topics & Assessment

Academic Track: Electrical and Electronics Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Transient Response of Second-Order RLC Circuits

" Series and parallel RLC transient response, Overdamped, critically damped, and underdamped response regimes,

Unit 4: Laplace Transform in Circuit Analysis

" Transformed circuit impedance models of R, L, and C, Initial and final value theorems, Complex frequency s-p

Unit 5: Two-Port Network Parameters

" Impedance Z, Admittance Y, Hybrid H, and Transmission ABCD parameters, Interconnection of two-port networks

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Find the Thevenin equivalent circuit across terminals containing both independent and voltage-controll
2. Solve for the transient current it in an underdamped series RLC circuit excited by a DC step voltage u
3. Derive the relationship between Z-parameters and ABCD transmission parameters for a reciprocal two-por

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* William H. Hayt, Jack E. Kemmerly - Engineering Circuit Analysis McGraw-Hill 8th Ed
* Charles K. Alexander, Matthew N.O. Sadiku - Fundamentals of Electric Circuits McGraw-Hill

Authorized NoteVault Educational Document " Verified for Registered University Students